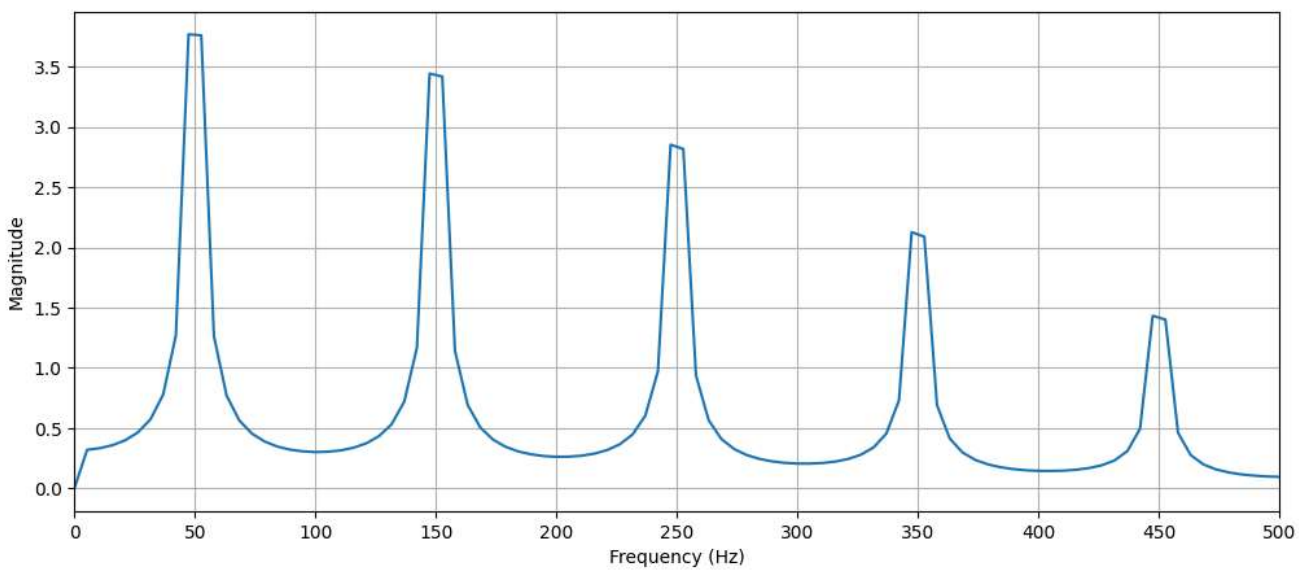
```
plt.grid(True)

plt.show()
```




```
fundamental = 50

orders = np.arange(1,21)

harmonics = []

for order in orders:

    target = order*fundamental

    idx = np.argmin(np.abs(freq-target))

    harmonics.append(magnitude[idx])

harmonics = np.array(harmonics)
```

```
for order, amp in zip(orders,harmonics):
    print(f"{order:2d}th Harmonic : {amp:.4f}")
```

```
 1th Harmonic : 3.7693
 2th Harmonic : 0.3004
 3th Harmonic : 3.4432
 4th Harmonic : 0.2609
 5th Harmonic : 2.8525
 6th Harmonic : 0.2046
 7th Harmonic : 2.1263
 8th Harmonic : 0.1442
 9th Harmonic : 1.4318
10th Harmonic : 0.0944
11th Harmonic : 0.9546
12th Harmonic : 0.0691
13th Harmonic : 0.7965
14th Harmonic : 0.0652
15th Harmonic : 0.7869
16th Harmonic : 0.0635
17th Harmonic : 0.7334
18th Harmonic : 0.0559
19th Harmonic : 0.6168
20th Harmonic : 0.0457
```

```
plt.figure(figsize=(10,4))

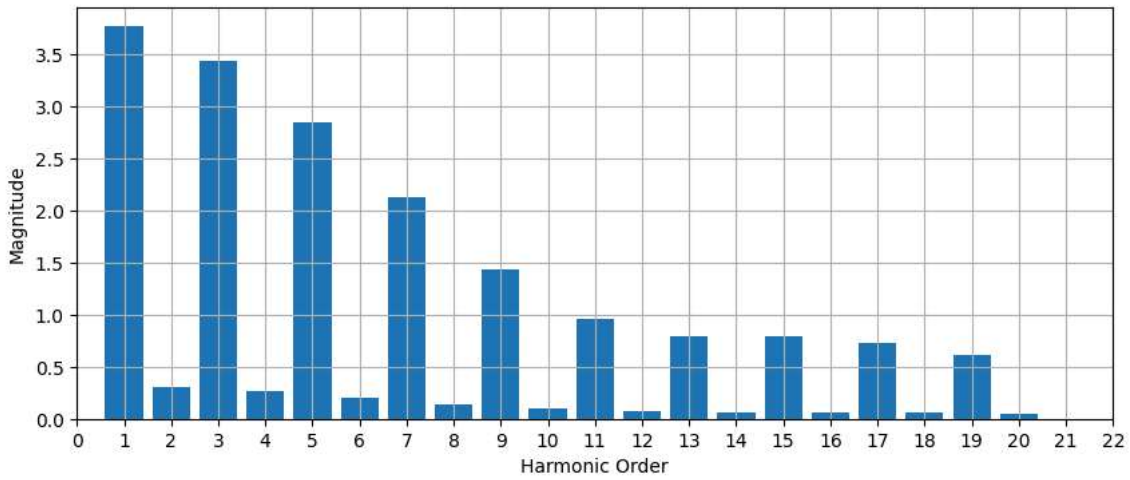
plt.bar(orders, harmonics)

plt.xlabel("Harmonic Order")
plt.xlim(0,22)
plt.xticks(np.arange(0,23,1))
```

```
plt.ylabel("Magnitude")

plt.grid(True)

plt.show()
```




```
V1 = harmonics[0]

THD = np.sqrt(np.sum(harmonics[1:]**2))/V1*100

print(f"THD = {THD:.2f}%")
```

```
THD = 145.06%
```